

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Information Technology) SEMESTER - 8 Summer 2025 (Regular)

Course :Bachelor of Technology (Information Technology) Branch : Engineering and Technology

Semester : SEMESTER - 8

Subject Code & Name: BTITC802 - MOBILE COMPUTING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1

In mobile computing, HLR is the abbreviation for which of the following?			
A) Home Live Register	B) House Live Register	C) Home Location Register	D) House Location Register
- 2

Which of the following is a type of wireless communication?			
A) LAN	B) WAN	C) PAN	D) All of the mentioned
- 3

The main disadvantage of digital communication is:			
A) Less security	B) More noise	C) High bandwidth requirement	D) Low data integrity
- 4

In 4G architecture, which component connects the user equipment (UE) to the EPC (Evolved Packet Core)?			
A) PDN Gateway	B) eNodeB	C) HSS	D) MME
- 5

What does SGSN stands for?			
A) Supporting Gateway Support Node	B) Supporting GGSN Support Node	C) Serving GPRS Support Node	D) Supporting GPRS Support Node
- 6

Which of the following components is unique to GPRS and not found in standard GSM?			
A) SGSN (Serving GPRS Support Node)	B) MSC (Mobile Switching Center)	C) BSC (Base Station Controller)	D) HLR (Home Location Register)
- 7

What is the main purpose of Mobile IP?			
A) Increase bandwidth	B) Maintain the same IP address while moving across networks	C) Assign new IP addresses to devices	D) Enhance data encryption

8	One disadvantage of I-TCP is:			
	A) It increases bandwidth	B) It breaks end-to-end TCP semantics	C) It uses too many IP addresses	D) It prevents data caching
9	In MANETs, each device acts as:			
	A) A router and host	B) Only a receiver	C) A server	D) A DNS resolver
10	Mobile IP uses which technique to forward packets from Home Agent to Mobile Node?			
	A) DNS resolution	B) Circuit switching	C) Broadcasting	D) Tunneling
11	What is data dissemination in mobile computing?			
	A) Encrypting user data	B) Process of distributing data to multiple mobile users	C) Uploading files to the cloud	D) Cleaning unnecessary data
12	Which mobile application commonly requires real-time data synchronization?			
	A) Calendar and email apps	B) Camera	C) Wallpaper changer	D) Voice recorder

Q2. Solve the following.

- | | |
|---|---|
| A) What are the benefits and applications of wireless mobile networks? | 6 |
| B) Explain the working principal of Digital Communication with the help of block diagram. | 6 |

Q3. Solve the following.

- | | |
|--|---|
| A) Explain the role of mobile computing in cloud-based services. | 6 |
| B) What are the different layers of mobile computing architecture? Explain with the help of diagram. | 6 |

Q4. Solve Any Two of the following.

- | | |
|---|---|
| A) Explain the role of various databases in GSM – HLR, VLR, EIR, and AuC. | 6 |
| B) Explain the working of LTE architecture from user equipment to the internet. | 6 |
| C) What are the various components of UMTS? Explain with a neat diagram. | 6 |

Q5. Solve Any Two of the following.

- | | |
|---|---|
| A) Explain the agent discovery and registration process of mobile IP with suitable diagram. | 6 |
| B) How is MANET different from traditional wireless networks? Explain in detail. | 6 |
| C) Explain why traditional TCP performs poorly over wireless networks and how Snooping TCP helps. | 6 |

Q6. Solve Any Two of the following.

- | | |
|---|---|
| A) Explain data dissemination broadcast model in mobile computing. | 6 |
| B) What do you understand by handover (handoff) management in mobile computing. Enlist and explain its various types. | 6 |
| C) What is data synchronization? Explain how synchronization is handled in mobile computing. | 6 |